



Examination
Final
Winter 2023
ECE 140 (Canizares)

Special Materials

Candidates may bring only the listed aids.

- Calculator - Scientific

Simple scientific calculators with up to four-line display are allowed in the exam. Calculators that are programmable, graphing or calculators with more than four-line display are not allowed.

Please print in pen:

Waterloo Student ID Number:

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WatIAM/Quest Login Userid:

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Times: Thursday 2023-04-20 at 16:00 to 18:30 (4 to 6:30PM)
Duration: 2 hours 30 minutes (150 minutes)
Exam ID: 5280943
Sections: ECE 140 LEC 001
Instructors: Claudio Canizares

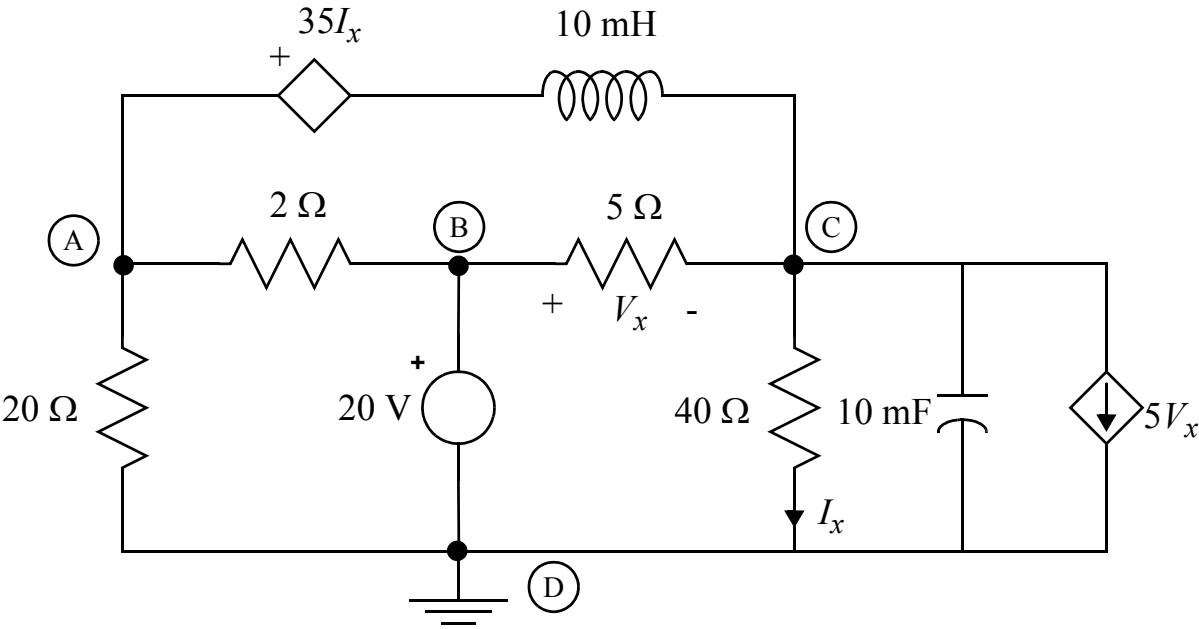
Instructions:

1. This is a CLOSED book examination.
2. Check that your name and student id is given correctly at the top of this page. This exam has assigned seating and you should sit only where the exam paper with your name and student id is placed.
3. Check that you have 5 questions in your paper. You should answer all 5 questions. The maximum marks for each question is given in the paper. Maximum marks for the entire exam is 50.
4. There are 16 pages in this exam. You can write on both sides of each sheet of paper. Return all pages at the end of the exam.
5. Each question has space below it (on the same page) and on the following page to write your answer. The two pages assigned to each question should be sufficient space to answer each question.
6. If you do not have enough space to answer a question on the two assigned pages, then continue writing where there is space and indicate where you have continued to answer this question. Pages 2, 13, 14 and 15 can be used as extra space to answer any question.
7. Clearly show all steps used in your solution. No marks will be given for numerical results unless accompanied with the correct solution method. Give units for all answers.
8. Simple scientific calculators with up to four-line display are allowed in the exam. Calculators that are programmable, graphing or calculators with more than four-line display are not allowed.
9. Other than the allowed calculator, you should not have any other electronic or communication devices (e.g. cell phone, laptop, etc.) with you while writing the exam. All such devices must be placed out of your reach during the exam.
10. Sign below to indicate you have read and understood all instructions.

Signature: _____

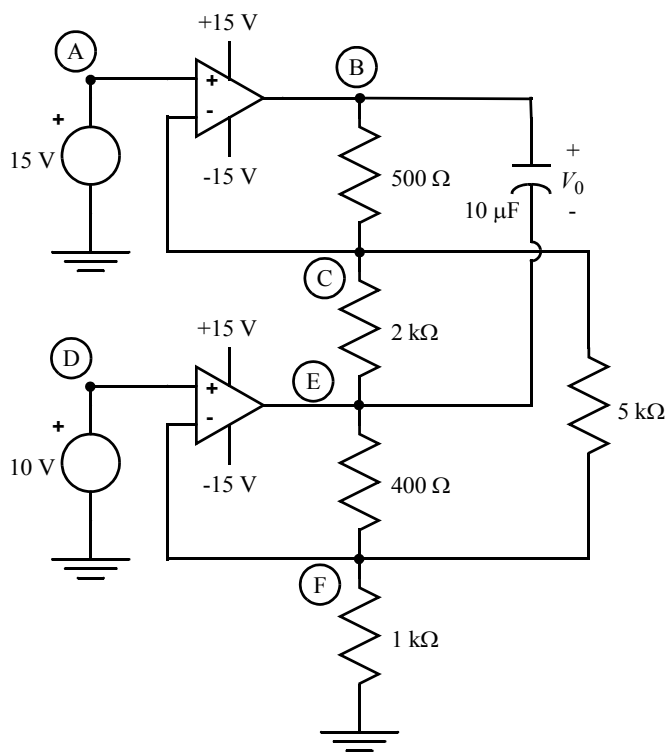
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Q1) Using *Nodal Analysis*, find the power supplied by the 20 V DC source in the following circuit. The circuit is operating in DC steady state. (10 marks)



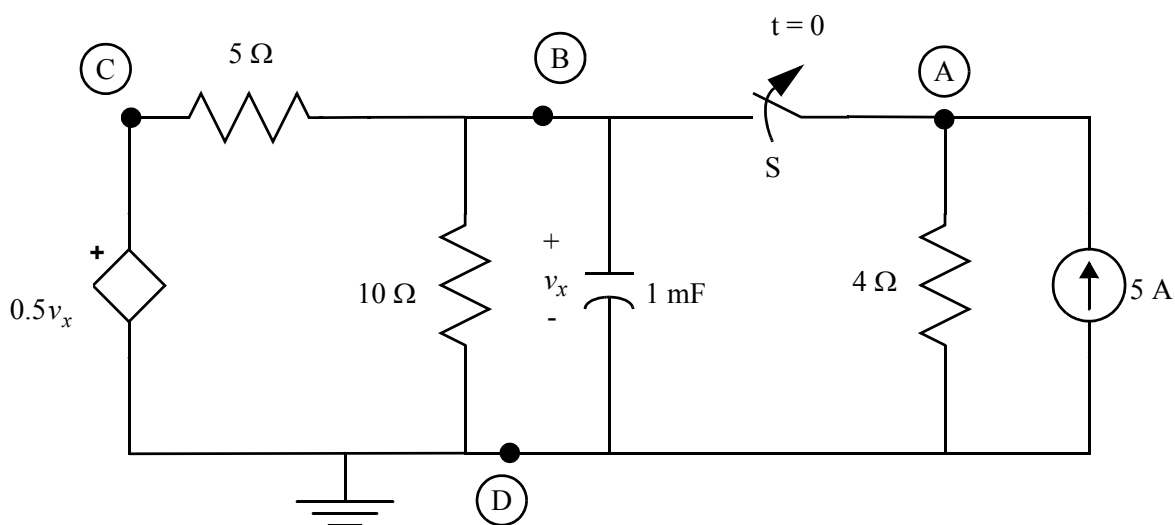
Assigned page to write answer for Q1.

Q2) Determine V_o the voltage across the capacitor in the following circuit operating in DC steady state. The OpAmps are non-ideal with the possibility of saturation, otherwise all values are ideal. (10 marks)



Assigned page to write answer for Q2.

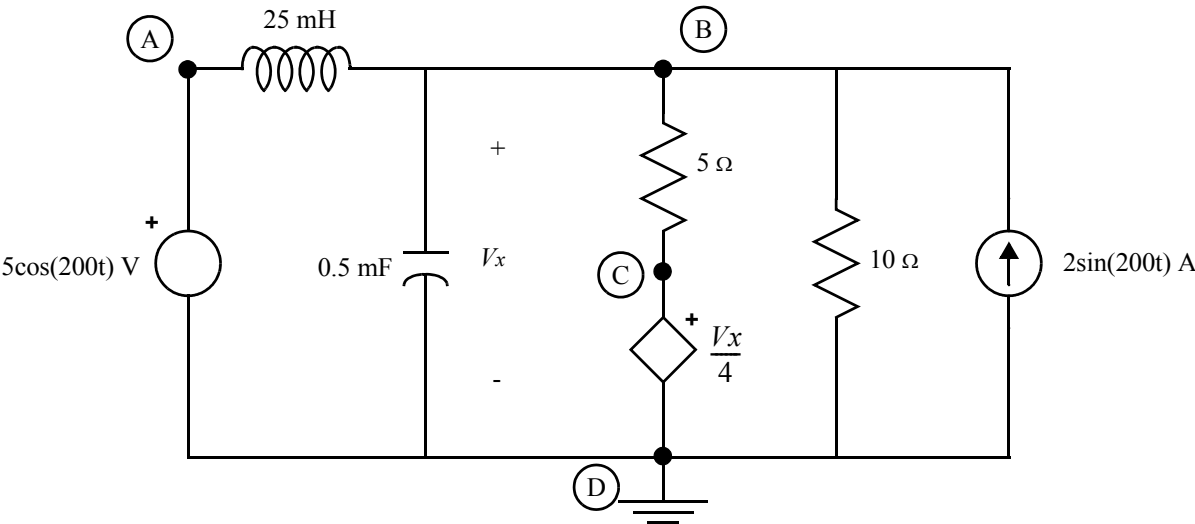
Q3) Calculate the voltage across the capacitor $v_x(t)$. The switch S is closed for a long time and the circuit reaches steady state for $t \leq 0$.



- (a) Find the initial condition on the capacitor $v_x(0)$. (2 marks)
- (b) At $t = 0$, switch S opens. Calculate an expression for the capacitor voltage $v_x(t)$ for $t \geq 0$. (8 marks)

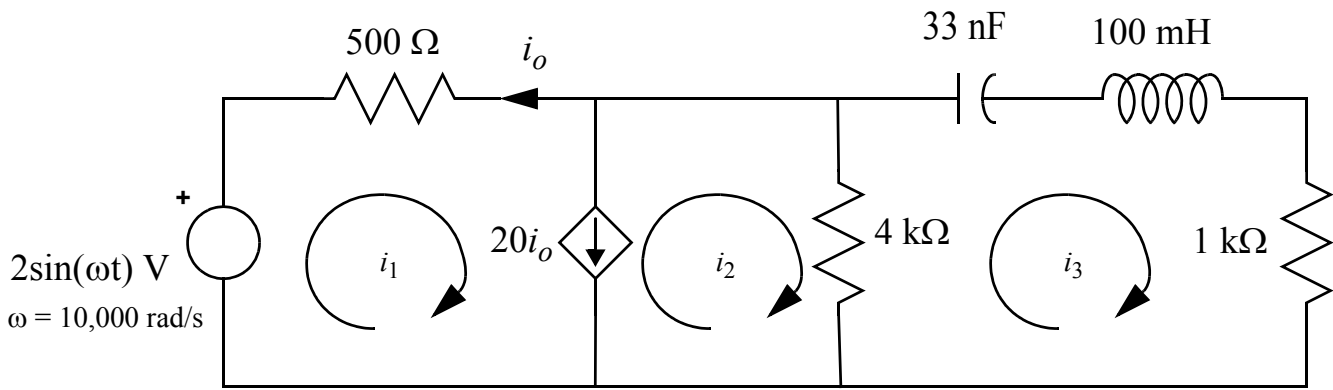
Assigned page to write answer for Q3.

Q4) The circuit below is operating in sinusoidal steady state. Calculate the voltage $V_B(t)$ using *Nodal Analysis*. Do not simplify the circuit in any way. (10 marks)



Assigned page to write answer for Q4.

Q5) The following circuit is operating in sinusoidal steady state at $\omega = 10,000 \text{ rad/s}$.



- (a) Use **Mesh Analysis** to solve for the mesh currents $i_1(t)$, $i_2(t)$, $i_3(t)$ in sinusoidal steady state. (8 marks)
- (b) Calculate the complex power supplied by the voltage source. (2 marks)

Assigned page to write answer for Q5.

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Formula Sheet

Current:	$i = \frac{dq}{dt}$	[A]	Ohm's Law: $v = Ri$ or $i = Gv$ where $G = \frac{1}{R}$
Voltage:	$v = \frac{dw}{dq}$	[V]	Resistors in Series: $R_{eq} = R_1 + R_2 + R_3$
Power:	$p = \frac{dw}{dt} = v \cdot i$	[W]	Resistors in Parallel: $\frac{1}{R_{eq}} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}$

Voltage, Current, and Energy

Device	Voltage	Current	Stored Energy
Resistor, R	$v(t) = Ri(t)$	$i(t) = \frac{v(t)}{R}$	
Inductor, L	$v(t) = L \frac{di(t)}{dt}$	$i(t) = i(t_o) + \frac{1}{L} \int_{t_o}^t v(t)dt$	$w(t) = \frac{L i(t)^2}{2}$
Capacitor, C	$v(t) = v(t_o) + \frac{1}{C} \int_{t_o}^t i(t)dt$	$i(t) = C \frac{dv(t)}{dt}$	$w(t) = \frac{C v(t)^2}{2}$

Transient Analysis

RC Circuit	$v(t) = v(\infty) + [v(t_0) - v(\infty)]e^{-\frac{t-t_0}{\tau}}; \tau = RC$
RL Circuit	$i(t) = i(\infty) + [i(t_0) - i(\infty)]e^{-\frac{t-t_0}{\tau}}; \tau = L/R$

AC Impedance

General	Resistor, R	Inductor, L	Capacitor, C
$Z = \frac{V}{I} = R + jX$ $Z \angle \theta_Z = \frac{V_m}{I_m} \angle \theta_V - \theta_I$	$R = R \angle 0^\circ$	$j\omega L = \omega L \angle 90^\circ$	$\frac{-j}{\omega C} = \frac{1}{\omega C} \angle -90^\circ$

AC Power

Complex, S [VA]	Real, P [W]	Reactive, Q [VAR]
$S = V_{rms} I_{rms}^* = P + jQ$	$P = V_{rms} I_{rms} \cos(\theta_V - \theta_I)$	$Q = V_{rms} I_{rms} \sin(\theta_V - \theta_I)$

$V_{rms} = \frac{V_m}{\sqrt{2}}; \quad I_{rms} = \frac{I_m}{\sqrt{2}}$

$R \angle \theta = R \cos \theta + jR \sin \theta; \quad a + jb = \sqrt{a^2 + b^2} \angle \tan^{-1} \left(\frac{b}{a} \right)$

$\sin(\theta) = \cos(\theta - 90^\circ); \quad -\sin(\theta) = \cos(\theta + 90^\circ); \quad -\cos(\theta) = \cos(\theta \pm 180^\circ)$